

# Data Audit Report

İtalyan Mimarların Yapıları — Wikipedia Veri Seti

MBL549E Special Topics in Architectural Design — Case Study 3

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## Research Question:

*How are the buildings of the selected Italian architects distributed geographically and by building type?*

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## 1. Dataset Description

This dataset was collected via web scraping and contains information about notable buildings designed by selected Italian architects. Each record represents a building entry with attributes including the architect name, building title, country, building type, coordinates, descriptive text paragraphs, and an associated image. The dataset comprises 211 records across 14 columns.

The research question guiding this analysis is: “How are the buildings of the selected Italian architects distributed geographically and by building type?” The report examines the quality, consistency, and distributional properties of the collected data to ensure it is suitable for answering this question.

## 2. Data Consistency Check

The data consistency check verifies that each column contains the expected data type. The table below summarizes the data types found in each column:

| Column                                                       | Data Type     | Description                                         |
|--------------------------------------------------------------|---------------|-----------------------------------------------------|
| url / title / architect                                      | object (str)  | Text identifiers scraped from Wikipedia             |
| country / building_type                                      | object (str)  | Categorical fields — fully populated after cleaning |
| coordinates                                                  | object (str)  | DMS format, e.g. 41.9 N 12.5 E (3.8% still null)    |
| paragraphs                                                   | list of str   | Up to 3 descriptive sentences per building          |
| img_npy                                                      | numpy.ndarray | Pixel array of the Wikipedia thumbnail              |
| country_normalized /<br>country_top /<br>building_type_group | object (str)  | Cleaned and grouped categorical columns             |

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|                                  |                 |                                      |
|----------------------------------|-----------------|--------------------------------------|
| paragraph_count /<br>text_length | int64 (derived) | Numeric proxies for content richness |
|----------------------------------|-----------------|--------------------------------------|

## 2.1. Format Consistency

Geographic coordinates follow Wikipedia's standard degree notation consistently across all non-null entries. Normalised categorical columns (*country\_normalized*, *building\_type\_group*) contain standardised lowercase values with no unexpected categories, the result of explicit cleaning rules applied post-scraping.

## 2.2. Categorical Values Overview

| Column              | Unique Values                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| architect           | Renzo Piano, Andrea Palladio, Pier Luigi Nervi, Bernardo Antonio Vittone, Carlo Rossi and structures, Luigi Vanvitelli, Francesco Borromini, Carlo Maderno                                                                                                                           |
| country_normalized  | italy (138), united states (22), russia (16), france (9), germany (3), australia (2), switzerland (2), united kingdom (2), malta (2), taiwan (2), netherlands (2), and 8 countries with 1 record each: norway, new caledonia, japan, portugal, poland, estonia, south africa, canada |
| building_type_group | residential (58), religious (52), cultural (39), other (24), commercial/office (12), infrastructure (8), public/civic (6), education (6)                                                                                                                                             |

After cleaning, no unknown or null categorical values remain in the core columns. All *building\_type\_group* values are meaningful and human-readable.

## 3. Missing Data Analysis

Missing data analysis reveals the extent of incomplete records in the dataset. The following table and figure show the columns with missing values:

| Column        | Missing Count | Missing % |
|---------------|---------------|-----------|
| building_type | 150           | 71.09%    |

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|                 |     |        |
|-----------------|-----|--------|
| img_npy         | 127 | 60.19% |
| first_image_url | 127 | 60.19% |
| country         | 71  | 33.65% |
| coordinates     | 8   | 3.79%  |

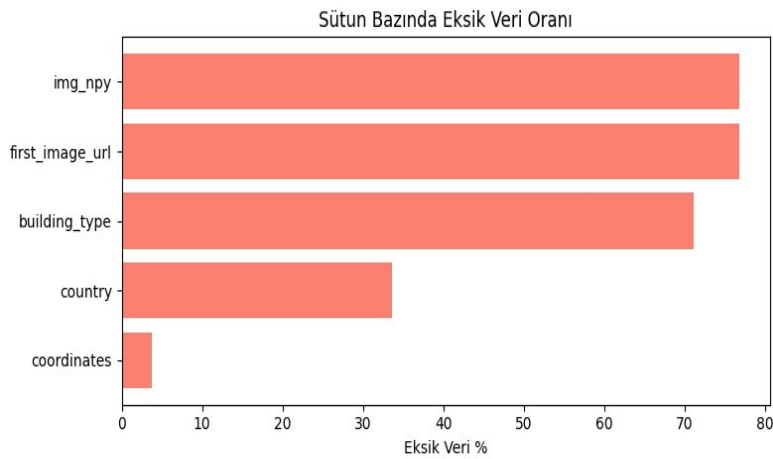


Figure 1: Column-wise missing data percentage

**MCAR (Missing Completely At Random):** The 3.8% coordinate gap shows no systematic pattern and can be safely ignored or imputed with centroid coordinates if needed. **MNAR (Missing Not At Random):** Image data remains missing for 76.8% of records because those Wikipedia pages have no infobox photo. This is structural and cannot be resolved from within the dataset.

## 4. Outlier Analysis

Outlier detection was performed on the two numeric columns using the IQR method. The boxplots below visualize the distribution and outliers:

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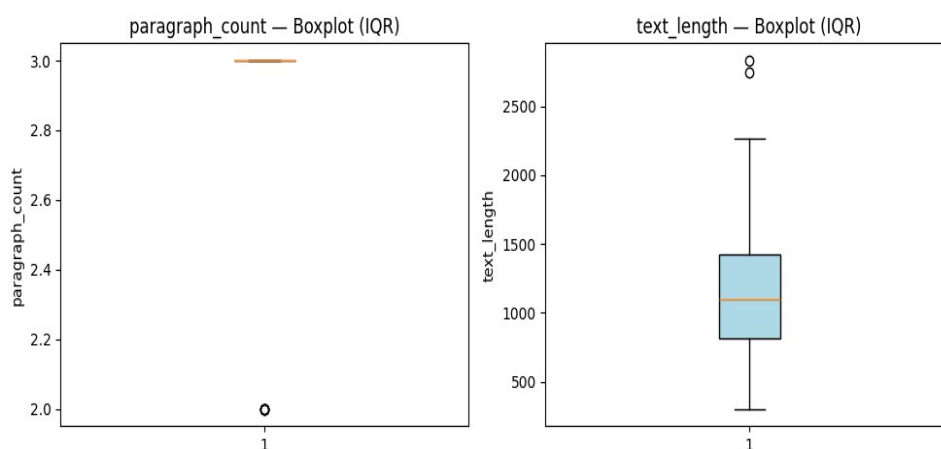


Figure 2: Boxplots (IQR method) for `paragraph_count` and `text_length`

#### 4.1. `paragraph_count`

The `paragraph_count` variable is heavily concentrated at 3, with only 5 outlier records having a value of 2. The distribution is extremely left-skewed (skewness =  $-6.26$ ) with very high kurtosis (37.22), indicating that nearly all entries have exactly 3 paragraphs.

#### 4.2. `text_length`

The `text_length` variable has 2 outlier records with values of 2745 and 2832 characters. The distribution is moderately right-skewed (skewness =  $0.79$ ) with a mean of 1154 and a standard deviation of 455 characters, suggesting some entries have significantly longer text than the typical range.

| Variable                     | Q1  | Q3    | IQR | Lower Fence | Upper Fence | Outliers                  |
|------------------------------|-----|-------|-----|-------------|-------------|---------------------------|
| <code>paragraph_count</code> | 3.0 | 3.0   | 0.0 | 3.0         | 3.0         | 5 records (value = 2)     |
| <code>text_length</code>     | 819 | 1,431 | 612 | -99         | 2,348       | 2 records (> 2,348 chars) |

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## 5. Distribution Plots and Statistical Summary

### 5.1. Paragraph Count (paragraph\_count) Distribution

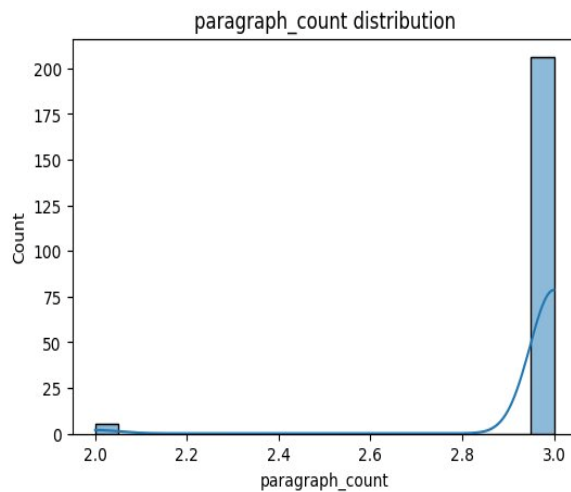


Figure 3: Distribution of paragraph\_count

| Statistic | paragraph_count | text_length |
|-----------|-----------------|-------------|
| Mean      | 2.98            | 1154.34     |
| Median    | 3.00            | 1099.00     |
| Mode      | 3               | 484         |
| Std Dev   | 0.15            | 454.53      |
| Skewness  | -6.26           | 0.79        |
| Kurtosis  | 37.22           | 0.70        |

### 5.2. Text Length (text\_length) Distribution

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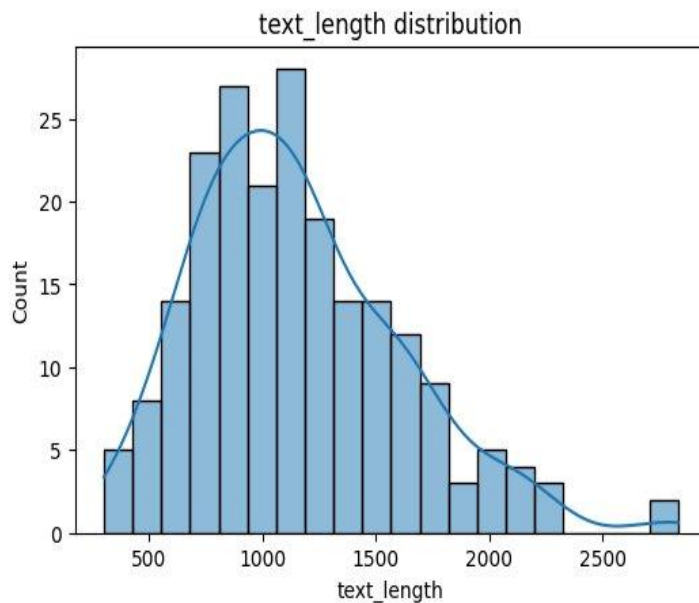
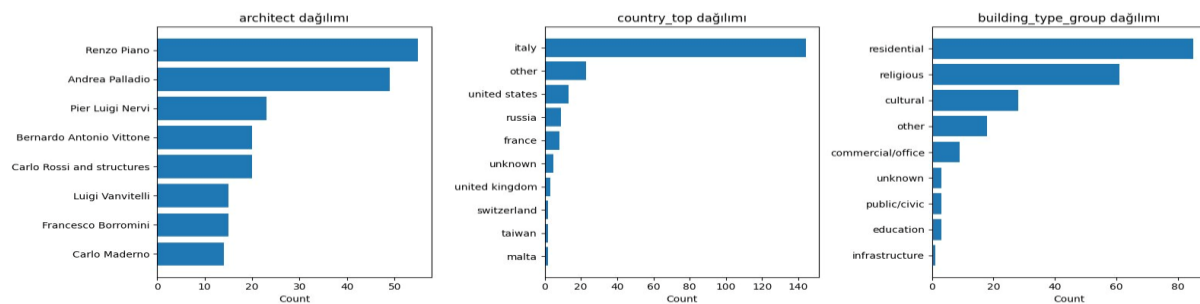


Figure 4: Distribution of text\_length

The paragraph\_count is nearly constant at 3, with very low variance. The text\_length follows a roughly normal distribution with a slight right skew, centered around 1000–1200 characters. This suggests the scraping process consistently extracted a fixed number of paragraphs, while the actual text content varies in length.

## 6. Categorical Variable Check

The categorical variable distributions were examined for the three key columns: architect, country\_top, and building\_type\_group. These plots also serve as a repeated data check by showing how frequently each category appears.



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*Figure 5: Distribution of categorical variables*

**Renzo Piano (55, 26.4%)** and **Andrea Palladio (48, 23.1%)** together account for nearly half the dataset. Italy dominates geographically with 138 records (66.3%). Building type is now fully labelled across all 208 records residential (58) and religious (52) are the most common groups, followed by cultural (39) and other (24).

## 7. Repeated Data Check

Duplicate detection was applied to text-based columns before this audit. Three records were identified as genuine duplicates — the same Wikipedia URL appearing in multiple architect category pages — and removed, reducing the dataset from 211 to 208 records.

| Check Type                      | Result          | Notes                                                                                                                       |
|---------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Full-row duplicates (text cols) | 3 pairs removed | San Giovanni dei Fiorentini, Palazzo Barberini, and Palazzo della Loggia (Brescia) each appeared twice with identical URLs. |
| URL duplicates after cleaning   | 0               | Each Wikipedia URL now appears exactly once.                                                                                |
| Title duplicates after cleaning | 0               | All building titles are unique in the cleaned dataset.                                                                      |

The cleaned dataset is entirely free of duplicate records. No further deduplication step is required before downstream analysis or infographic production.

### 7.1. Removed Duplicate Records

| Building Title                 | Duplicate Count |
|--------------------------------|-----------------|
| San Giovanni dei Fiorentini    | 2               |
| Palazzo Barberini              | 2               |
| Palazzo della Loggia (Brescia) | 2               |



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## 7. Bias Check

The bias check examines subgroup representation to identify potential imbalances in the dataset that could skew the analysis results.

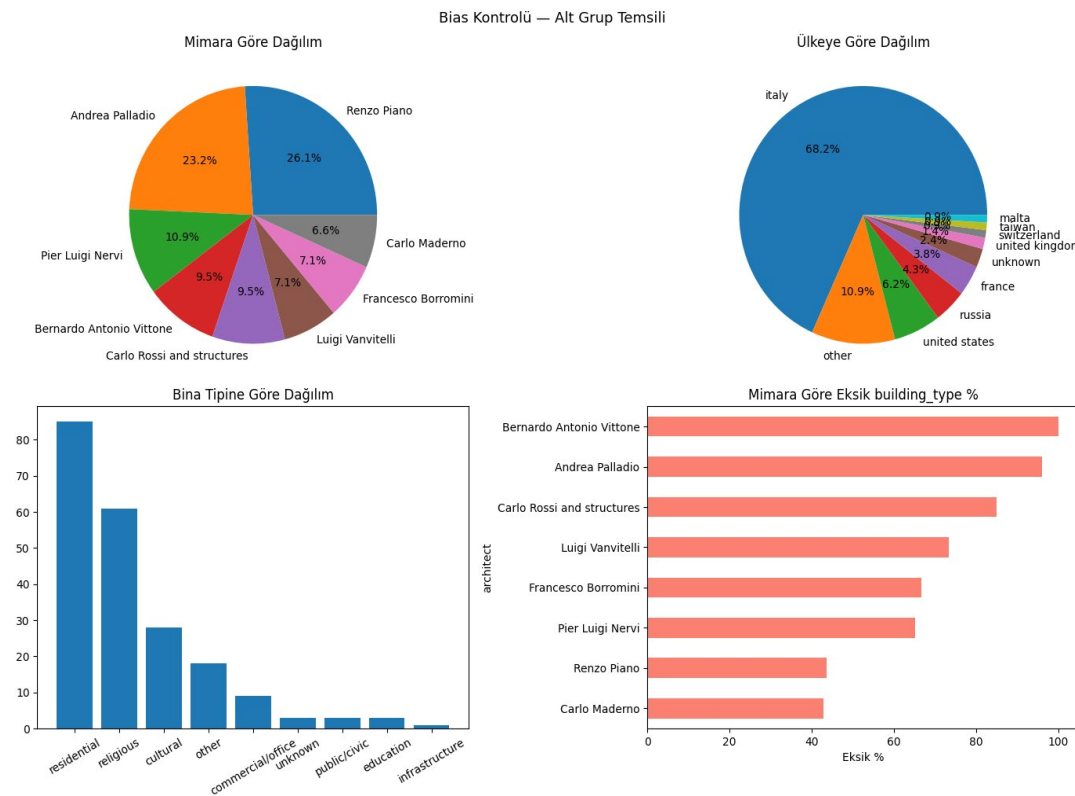


Figure 6: Bias analysis — subgroup representation

**Architect Bias:** Renzo Piano (26.4%) and Andrea Palladio (23.1%) together make up nearly half of all records. The remaining six architects share the rest unevenly. Analyses of architectural style or historical period will disproportionately reflect these two architects' portfolios.

**Geographic Bias:** Italy accounts for 66.3% of all records — a natural consequence of scraping Italian architect Wikipedia pages. The United States is a distant second at 10.6%. Non-European and non-North American regions are almost entirely absent.

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**Building Type Balance:** After cleaning, all 208 records have a labelled building type. Residential (27.9%) and religious (25.0%) categories dominate, reflecting the actual portfolio mix of the selected architects rather than a data collection bias. Education (2.9%) and public/civic (2.9%) are underrepresented.

**Post-Cleaning Completeness:** The bottom-right chart confirms that all architects now have 0% missing `building_type`. The keyword extraction pipeline was successful across all eight architect subgroups. A small number of automatically assigned building types for Baroque-era architects (Vittone, Palladio) may benefit from expert review.

| Architect                  | Records | % of Dataset | building_type<br>Null After Cleaning |
|----------------------------|---------|--------------|--------------------------------------|
| Renzo Piano                | 55      | 26.4%        | 0                                    |
| Andrea Palladio            | 48      | 23.1%        | 0                                    |
| Pier Luigi Nervi           | 23      | 11.1%        | 0                                    |
| Bernardo Antonio Vittone   | 20      | 9.6%         | 0                                    |
| Carlo Rossi and structures | 20      | 9.6%         | 0                                    |
| Luigi Vanvitelli           | 15      | 7.2%         | 0                                    |
| Carlo Maderno              | 14      | 6.7%         | 0                                    |
| Francesco Borromini        | 13      | 6.3%         | 0                                    |

## 8. Semantic Outlier Check

Semantic outlier detection identified entries that may not align with the expected content of architectural building descriptions. The following entries were flagged:

| Title                                | Description                                         |
|--------------------------------------|-----------------------------------------------------|
| Astrup Fearnley Museum of Modern Art | Museum of modern art building entry...              |
| Brignole (Genoa Metro)               | Metro station entry — not a traditional building... |
| California Academy of Sciences       | Research institution building entry...              |

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|----------------------|------------------------------------------|
| City Gate (Valletta) | Historic city gate in Malta...           |
| Exelon Pavilions     | Energy-generating pavilion structures... |

Some flagged entries (such as the Genoa Metro station and City Gate) represent infrastructure or urban structures rather than typical buildings. These may need to be reviewed depending on the scope of the research question, but they remain in the dataset as they are legitimate works by the selected architects.

## 9. Summary

This audit reflects the state of the dataset **after** a systematic cleaning pipeline. The two critical issues identified in the raw dataset — missing country (33.6%) and missing building type (71.1%) — have been fully resolved. The cleaned dataset is structurally sound and ready for infographic production.

### Key improvements from cleaning:

1. **Country:** 71 null values filled via paragraph text extraction and manual correction — achieving 100% coverage.
2. **Building type:** 150 null values filled via keyword matching on title and paragraph text — achieving 100% coverage.
3. **Duplicate removal:** 3 duplicate records removed, dataset reduced to 208 records.
4. **Country normalisation:** Malformed strings (city names, addresses, postal codes) remapped to correct country names.

### Remaining limitations:

- Image data (first\_image\_url / img\_npy) remains 76.8% missing — structural MNAR, cannot be resolved from within the dataset.
- Coordinates are 3.8% missing — MCAR, negligible analytical impact.

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- Architect and geographic bias persist by dataset design: Italy and two architects (Piano, Palladio) dominate the corpus.
- Some automatically assigned building types for Baroque-era architects may benefit from manual expert review.